

IMPERIAL COLLEGE LONDON · GROUP PROJECT · JUNE 2025

The Kelvin Water Dropper

Overview

This project investigated how a Kelvin water dropper builds charge through electrostatic induction, and how environmental conditions affect its charging rate.

What I did

Working in a group of four, I helped build a theoretical model connecting air humidity to charging rate and tested the prediction against experimental measurements.

Analysis and communication

The comparison used regression and uncertainty propagation. We presented the result in a ten-minute group video; this webpage and its PDF preserve the public project summary.

ELECTROSTATICS

MODELLING

UNCERTAINTY ANALYSIS